FIITJEE

ALL INDIA INTEGRATED TEST SERIES

PART TEST – III

JEE (Advanced)-2021

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **One Section**: **Section-A.**

1. ***Section–A (01 – 07, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (08– 14, 26 – 32, 44 – 50) *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*

   Partial Marks **+1** for each correct option provided no incorrect option is selected.

   Section–A (15 – 18, 33 – 36, 51 – 54) *contains 6 paragraphs. Each paragraph is having 2 multiple choice questions having 4 options with* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

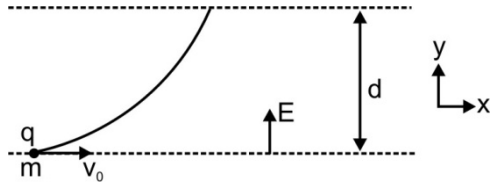
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

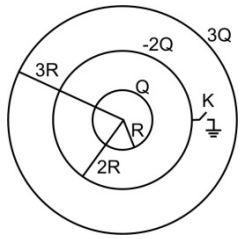
1. Charge q of mass m is projected with velocity v0 along x-axis in uniform electric field E along y-axis. Radius of curvature of charge when it has travelled distance d along y-axis is ? Neglect gravity and it is given that $\frac{qEd}{m} = 2v_0^2$



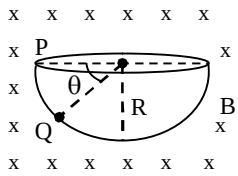
(A) $\frac{v_0^2 m}{qE}\sqrt{5}$ (B) $\frac{v_0^2 m}{2qE}\sqrt{5}$

(C) $\frac{5}{2}\frac{v_0^2 m}{qE}\sqrt{5}$ (D) $\frac{5v_0^2 m}{qE}\sqrt{5}$

2. Find charge flown from earth when K is switch on.
(A) –Q (B) Q
(C) 2Q (D) –3Q



3. A charged sphere of mass m and charge – q starts sliding along the surface of a smooth hemispherical bowl, at position P. The region has a transverse uniform magnetic field B. Normal force by the surface of bowl on the sphere at position Q is:



(A) $mg \sin\theta + qB\sqrt{2gR\sin\theta}$

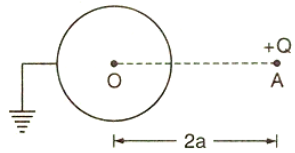
(B) $3mg \sin\theta + qB\sqrt{2gR\sin\theta}$

(C) $mg \sin\theta - qB\sqrt{2gR\sin\theta}$

(D) $3mg \sin\theta - qB\sqrt{2gR\sin\theta}$

Space for Rough work

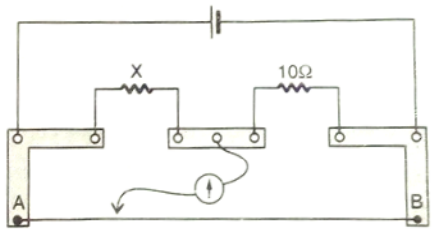
4. A point charge Q is placed at a distance 2a from the center of a thin conducting uncharged spherical shell of radius 'a' as shown in the figure. Then, the charge on the shell after connecting to the earth, will be



(A) $-\frac{Q}{2}$ (B) Q

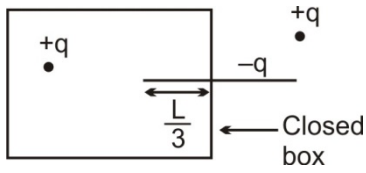
(C) –Q (D) Zero

5. A meter bridge is set-up as shown, to determine an unknown resistance 'X' using a standard 10 ohm resistor. The galvanometer shows null point when tapping-key is at 52 cm mark. The end-corrections are 1 cm and 2 cm respectively for the ends A and B. The determined value of 'X' is:



(A) 10.2 ohm
(B) 10.6 ohm
(C) 10.8 ohm
(D) 11.1 ohm

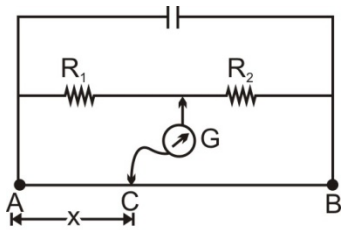
6. Figure shows two point charges +q and +q and a metal rod of length L carrying a charge –q with a part L/3 of its length inside a box. The electric flux through the box is



(A) $\frac{q}{3\varepsilon_0}$ (B) $\frac{4q}{3\varepsilon_0}$

(C) $\frac{2q}{3\varepsilon_0}$ (D) $\frac{5q}{3\varepsilon_0}$

7. In the meter bridge experiment shown in figure, the balance length AC corresponding to null deflection of the galvanometer is x. What would be the balance length if the radius of the wire AB is doubled?



(A) $\frac{x}{2}$ (B) x

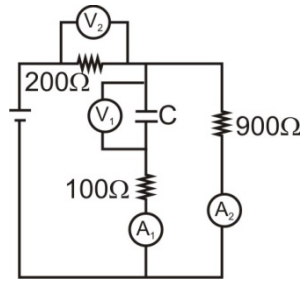
(C) 2x (D) 4x

Space for Rough work

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

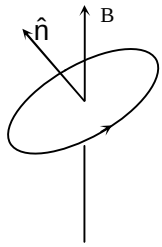
8. An electrical circuit is shown in the given figure. The resistance of each voltmeter is infinite and each ammeter is 100Ω. The charge on the capacitor of 100 μF in steady state is 4mC. Choose correct statements(s) regarding the given circuit



(A) Reading of voltmeter V_2 is 8 V
(B) Reading of ammeter A_1 is zero and A_2 is 1/25 A
(C) Reading of voltmeter V_1 is 40 V
(D) EMF of the ideal cell is 66V

9. A particle of mass 2 kg and charge 1 mC is projected vertically with a velocity 10 ms^{-1}. There is a uniform horizontal electric field of 10^4 N/C
(A) The horizontal range of the particle is 10 m
(B) the time of flight of the particle is 2 sec
(C) the maximum height reached is 5 m
(D) the horizontal range of the particle is 5 m

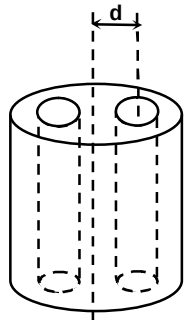
10. An electron in the ground state of hydrogen atom is revolving in anticlockwise direction in a circular orbit of radius R. The atom is placed in a uniform magnetic field $\vec{B}$ such that the plane normal of the electron orbit makes an angle of 30° with the magnetic induction. (The symbols have their usual meaning)



(A) The orbital magnetic dipole moment of the electron is $\frac{eh}{4\pi m_e}$
(B) The orbital magnetic dipole moment of the electron is $\frac{eh}{\pi m_e}$
(C) The torque experienced by the orbiting electron is $\frac{ehB}{8\pi m_e}$
(D) The torque experienced by the orbiting electron is $\frac{ehB}{4\pi m_e}$

Space for Rough work

11. An infinitely long cylindrical conductor of radius R contains current of uniform density J along axis of the cylinder. Two infinitely long cylindrical holes of radius r are drilled symmetrically throughout the length of the cylinder. The axes of the holes are parallel to the axis of cylinder and at distance d from it in the same plane.



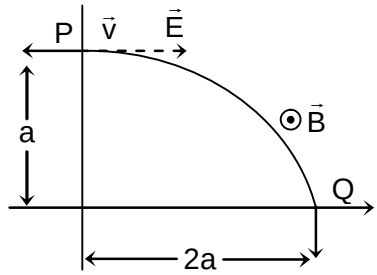
(A) Magnetic field on the axis of one of the hole will be $\frac{\mu_0 J}{2}\left(d-\frac{r^2}{2d}\right)$

(B) Magnetic field on the axis of one of the hole will be $\frac{\mu_0 J}{2}\left(d-\frac{r^2}{d}\right)$

(C) Magnetic field on the axis of cylinder will be $\frac{\mu_0 J}{2}\frac{r^2}{d}$

(D) Magnetic field on the axis of cylinder will be zero

12. A particle of charge + q and mass m moving under the influence of a uniform electric field $E\hat{i}$ and uniform magnetic field $B\hat{k}$ follows a trajectory from P to Q as shown in the figure. The velocities at P and Q are $v\hat{i}$ and $-2v\hat{j}$ which of the following statement (s) is/are correct ?



(A) $E=\frac{3}{4}\left[\frac{mv^2}{qa}\right]$

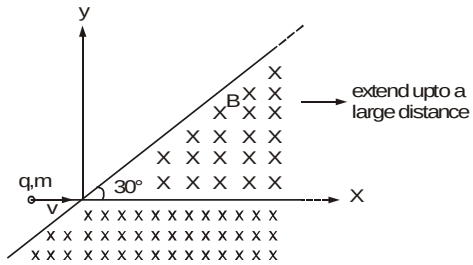
(B) Rate of work done by the electric field at P is $\frac{3}{4}\left[\frac{mv^3}{a}\right]$

(C) Rate of work done by the electric field at P is zero

(D) Rate of work done by both the fields at Q is zero

Space for Rough work

13. A charge particle of charge q, mass m is moving with initial velocity 'v' as shown in figure in a uniform magnetic field –. Select the correct alternative/alternatives:



(A) Velocity of particle when it comes out from magnetic field is $\vec{v}=v\cos 60^o\,\vec{i}-v\sin 60^o\,\hat{j}$

(B) Time for which the particle was in magnetic field is $\frac{\pi m}{3qB}$

(C) Distance travelled in magnetic field is $\frac{\pi mV}{3qB}$

(D) The particle will never come out of magnetic field

14. A long straight conductor of circular cross-section of radius R carries a current i_o whose current density J is assumed to be varying with radial distance r as $J = J_o\left(\frac{r}{R}\right)$, where J_o is a positive constant, then

(A) Magnetic field outside varies as $\frac{J_o\mu_o r^2}{3R}$

(B) Magnetic field inside varies as $\frac{\mu_o J_o R^2}{3r}$

(C) Inside the conductor graph between magnetic field and radial distance is parabolic

(D) outside the conductor graph between magnetic field and radial distance is hyperbolic

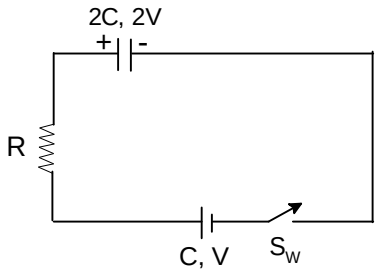
Space for Rough work

Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 15 and 16

Two capacitors of capacitance is 2 C and C are charged to potential difference 2V and V respectively. Now the charged capacitors are connected with a resistor R as shown in figure. Switch is closed at t = 0



15. What is potential difference across each capacitor in steady state?

(A) $\frac{2V}{5}$ (B) $\frac{2}{3}V$

(C) $\frac{5V}{3}$ (D) $\frac{3}{5}V$

16. What is current in circuit as function of time?

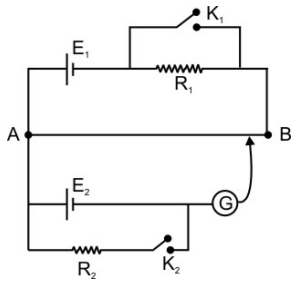
(A) $\frac{V}{R}e^{-3t/RC}$ (B) $\frac{2V}{R}e^{-2t/3RC}$

(C) $\frac{V}{3R}e^{-t/2RC}$ (D) $\frac{V}{R}e^{-3t/2RC}$

Space for Rough work

Paragraph for Question Nos. 17 and 18

Figure shows the circuit of a potentiometer. The length of the potentiometer wire AB is 50 cm. The EMF E1 of the battery is 4 V, having negligible internal resistance. Value of R1 and R2 are 15Ω and 5Ω, respectively. When both the keys are open, the null point is obtained at a distance of 31.25 cm from A, but when both the keys are closed, the balance length reduces to 5 cm only. Given $R_{AB} = 10\Omega$



17. The internal resistance of the cell E_2 is
(A) 4.5 Ω (B) 5.5 Ω
(C) 6.5 Ω (D) 7.5 Ω

18. The balance length when key K_2 is open and K_1 is closed is given by
(A) 10.5 cm (B) 11.5 cm
(C) 12.5 cm (D) 13.5 cm

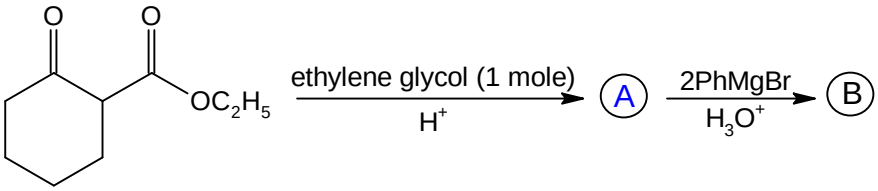
Space for Rough work

Chemistry

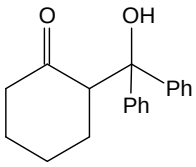
PART – II

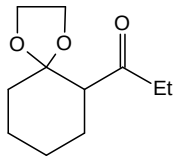
SECTION – A
(One Options Correct Type)

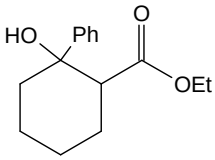
This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

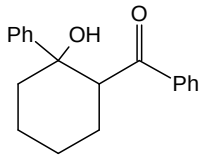
19. 

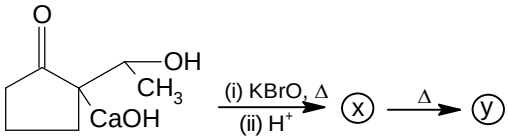
Product B of the reaction is

(A) 

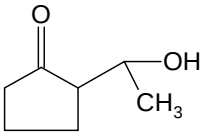
(B) 

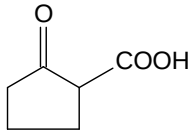
(C) 

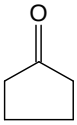
(D) 

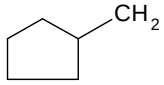
20. 

Product y of the above reaction is

(A) 

(B) 

(C) 

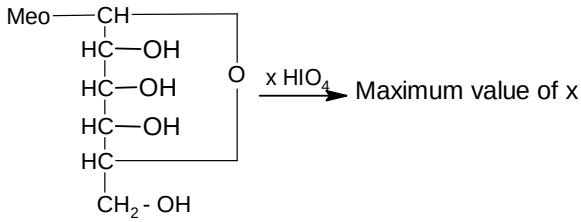
(D) 

Space for Rough work

21. $Ph-COOH$ *and* $Ph-CH_3$ can be separated by

(A) $NaHCO_3$ (B) $Aq.\ NaHCO_3 + n-hexane$

(C) H_2O (D) $n-Hexane$

22. 

Maximum value of x

(A) 1 (B) 2

(C) 3 (D) 4

23. Which of following amino acid has lowest iso-electric point.

(A) Glycine (B) Alanine

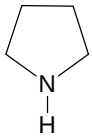
(C) Aspartic acid (D) Lysine

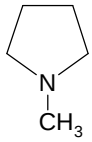
24. Acetaldehyde and Banzaldehyde can be differentiated by

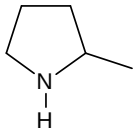
(A) Fenling test (B) Iodoform test

(C) Iodoform test (D) both A & B

25. $Br-CH_2-(CH_2)_2-CH_2-Br+CH_3-NH_2 \rightarrow$ Product of reaction is

(A) 

(B) 

(C) 

(D) None

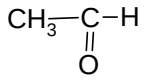
Space for Rough work

(One or More than one correct type)

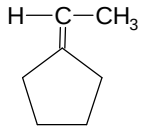
This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

26. Which of the following are disaccharides ?
(A) Glucose (B) Cane sugar
(C) Maltose (D) Starch

27. A 3° optically active alcohol $C_9H_{18}O(A)$, on dehydration with conc. H_2SO_4 produces B (C_9H_{16}) which exists in two stereo isomeric forms. For ozonolysis of B followed by work up with $Zn-H_2O$ produces

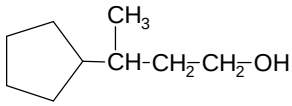


and C $(C_7H_{12}O)$. C on treatment with $LiAlH_4$ produces D $(C_7H_{14}O)$. D on dehydration produced

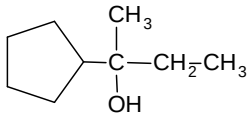


Identify correct answer.

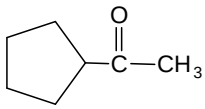
(A) A is



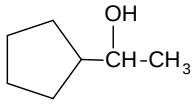
(B) A is



(C) D is

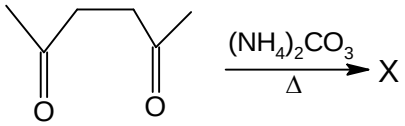


(D) D is



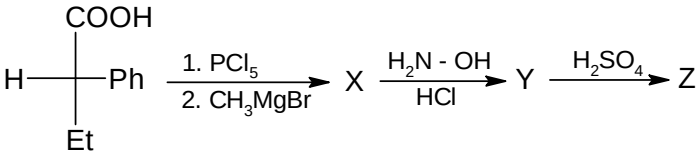
28. Which of the following statement is/are correct?
(A) Cellulose nitrate is a natural polymer
(B) Vulcanization increases rigidity by restricting the movement of the polymer chain
(C) HDPE is a linear polymer where as L.D.P.E is a branched polymer
(D) Cellulose is an example of an unbranched polysaccharide

Space for Rough work

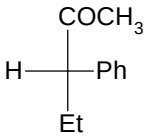
29. 

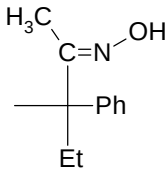
Which of the following statement is correct?

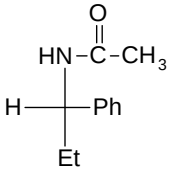
(A) Product x is 5 member ring
(B) Product x is aromatic
(C) Product x is anti aromatic
(D) Product x has a 6 member ring

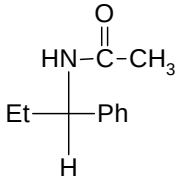
30. 

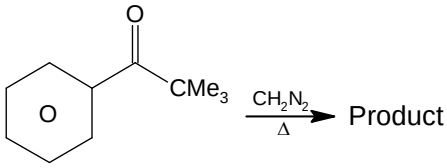
X, Y and Z are

(A) 

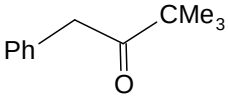
(B) 

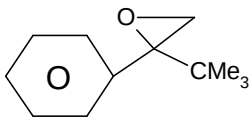
(C) 

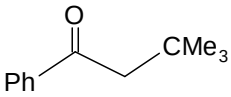
(D) 

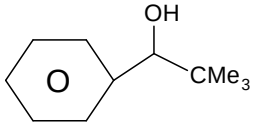
31. 

The possible product can be

(A) 

(B) 

(C) 

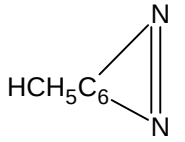
(D) 

Space for Rough work

32. $C_6H_5-CH_2-I \xrightarrow[\Delta]{NaN_3}$ product s.

Reaction is assumed to involve nitrene as inter mediate then various possible product are

(A) $C_6H_5CH_2NH_2$ (B) $C_6H_5N=CH_2$

(C) $C_6H_5CH=NH$ (D) 

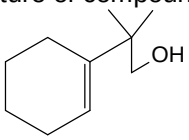
Paragraph Type (One Option Correct)

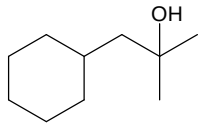
This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

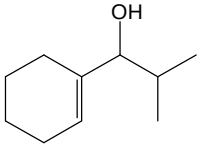
Paragraph for Question Nos. 33 and 34

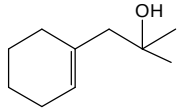
$C_7H_{11}Br \xrightarrow[(ii)\ H_3O^+]{(i)\ CH_3-\overset{\overset{O}{\|}}{C}-CH_3} (A) \xrightarrow[Et_2O]{Mg} (B) \xrightarrow[Heat]{H_2SO_4} C_{10}H_{16} \ (C) \xrightarrow[Pt,\ D]{H_2}$ 2 –Methyl Propyl – Cyclohexane

33. Structure of compound B is

(A) 

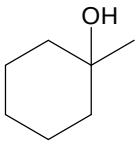
(B) 

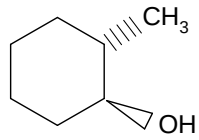
(C) 

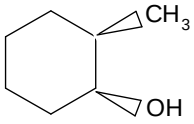
(D) 

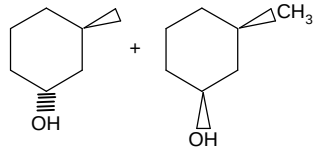
34. $(A) \xrightarrow[H_3O^+]{(i)\ LiAlH_4} \xrightarrow[H_2O_2,\ NaOH]{(i)\ B_2H_6\ /\ THF} X$

X is

(A) 

(B) 

(C) 

(D) 

Space for Rough work

Paragraph for Question Nos. 35 and 36

N, S, Halogens and Phosphorous present in Organic compound are detected by Lassaigne test by fusing the compound with Na metal. The compound formed are extracted from fused by boiling it with distilled water. The extract is known as Sodium fusion extract or Lassaigne's extract.

35. Violet colour in the detection of Sulphur with Sodium nitroprosside is due to formation of

(A) $Na_3\left[Fe(CN)_5 NoS\right]$ (B) $Na_4\left[Fe(CN)_5 NoS\right]$

(C) $Na_4\left[Fe(CN)_5\right]$ (D) $Na_2\left[Fe(CN)_5 S\right]$

36. An Organic compound containing N, S and O as extra element is fused with metallic sodium and then extracted with distilled water, which species is not present in the Lassaigne's extract.

(A) $NO_3^{\ominus}$ (B) $CN^{\ominus}$ (C) $\bar{S}CN$ (D) S^{2-}

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. If the function $f(x) = -4e^{\frac{1-x}{2}} + 1 + x + \frac{x^2}{2} + \frac{x^3}{3}$ and $g(x) = f^{-1}(x)$ then value of $g'\left(\frac{-7}{6}\right)$ equals

(A) $\frac{1}{5}$ (B) $-\frac{1}{5}$

(C) $\frac{6}{7}$ (D) $-\frac{6}{7}$

38. If $\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}$, then $\int_0^{\infty} \frac{\sin^3 x}{x} dx$ is equal to

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$ (D) $\frac{3\pi}{2}$

39. The fundamental period of the function
$f(x) = c^{\sin^2 x + \sin^2\left(x+\frac{\pi}{3}\right)} + \cos x \cos\left(x + \frac{\pi}{3}\right)$ is (where c is constant)

(A) 1 (B) $\frac{\pi}{2}$

(C) π (D) None of these

Space for Rough work

40. A function f(x) defined as
$f(x)=\begin{cases}\sin x, & x \text{ is rational}\\ \cos x, & x \text{ is irrational}\end{cases}$ is continuous at

(A) $x=n\pi+\frac{\pi}{4}, n\in I$ (B) $x=n\pi+\frac{\pi}{8}, n\in I$

(C) $x=n\pi+\frac{\pi}{6}, n\in I$ (D) $x=n\pi+\frac{\pi}{3}, n\in I$

41. If $f(x)=\begin{cases}e^{x^2+x}, & x>0\\ ax+b, & x\le 0\end{cases}$ is differentiable at x = 0, then

(A) $a=1, b=-1$ (B) $a=-1, b=1$

(C) $a=1, b=1$ (D) $a=-1, b=-1$

42. The angle of intersection of the normals at the point $\left(-\frac{5}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$ of the curves $x^2-y^2=8$ and $9x^2+25y^2=225$ is

(A) 0 (B) $\frac{\pi}{2}$

(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

43. Let f be a continuous and differentiable function. If the tangent to $y=f(x)$ at x = a also the normal to y = f(x) at x = b, then there exists at least one $c\in(a, b)$ such that

(A) $f'(c)=0$ (B) $f'(c)>0$

(C) $f'(c)<0$ (D) None of these

Space for Rough work

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

44. The function $\frac{\sin(x+a)}{\sin(x+b)}$ has no maxima or minima if

(A) $b-a=n\pi, n\in I$ (B) $b-a=(2n+1)\pi, n\in I$

(C) $b-a=2n\pi, n\in I$ (D) None of these

45. The abscissa of a point on the curve $xy=(a+x)^2$, the normal at which cuts off numerically equal intercepts from the coordinate axes, is

(A) $-\frac{a}{\sqrt{2}}$ (B) $\sqrt{2}a$

(C) $\frac{a}{\sqrt{2}}$ (D) $-\sqrt{2}a$

46. Which of the following function has point of extremum at x = 0?

(A) $f(x)=e^{-|x|}$

(B) $f(x)=\sin|x|$

(C) $f(x)=\begin{cases}x^2+4x+3, x<0\\ -x, x\geq 0\end{cases}$

(D) $f(x)\begin{cases}|x|, x<0\\ \{x\}, 0\leq x<1\end{cases}$ (where $\{x\}$ represents fractional part function).

47. Let $f:R\to R$ be any function and $g(x)=\frac{1}{f(x)}$. Then which of the following is/are not true?

(A) g is onto if f is onto (B) g is one – one if f is one to one

(C) g is continuous if f is continuous (D) g is differentiable if f is differentiable

Space for Rough work

48. Let $f(x) = \sin x + ax + b$. Then which of the following is/are true?
(A) f(x) = 0 has only one real root which is positive if a > 1, b < 0.
(B) f(x) = 0 has only one real root which is negative if a > 1, b > 0.
(C) f(x) = 0 has only one real root which is negative if a < -1, b < 0.
(D) None of these

49. Let $g'(x) > 0$ and $f'(x) < 0 \forall x \in R$. Then
(A) $g(f(x+1)) > g(f(x-1))$ (B) $f(g(x-1)) > f(g+1)$
(C) $g(f(x+1)) < g(f(x-1))$ (D) $g(g(x+1)) < g(g(x-1))$

50. If $f(x) = \int_0^x (\cos(\sin t) + \cos(\cos t))dt$, then $f(x+\pi)$ is
(A) $f(x) + f(\pi)$ (B) $f(x) + 2f(\pi)$
(C) $f(x) + f\left(\frac{\pi}{2}\right)$ (D) $f(x) + 2f\left(\frac{\pi}{2}\right)$

Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 51 and 52

If $L = \lim_{x\to\infty}\left(x+1-\sqrt{ax^2+x+3}\right)$ exists finitely then

51. The value of a is
(A) -1 (B) $\frac{1}{2}$
(C) 1 (D) $\frac{3}{2}$

Space for Rough work

52. The value of L is

(A) $\frac{1}{3}$ (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) $\frac{2}{3}$

Paragraph for Question Nos. 53 and 54

$y = f(x)$ satisfies the relation $\int_2^x f(t)dt = \frac{x^2}{2} + \int_x^2 t^2 f(t)dt.$

53. The range of y = f(x) is

(A) $[0, \infty)$ (B) R

(C) $(-\infty, 0]$ (D) $\left[-\frac{1}{2}, \frac{1}{2}\right]$

54. The value of $\int_{-2}^{2} f(x)dx$ is

(A) 0 (B) –2

(C) $2\log_e 2$ (D) None of these

Space for Rough work